MIL-P-11824C(MU)
7 October 1970
SUPERSEDING
MIL-P-11824B(MU)
16 April 1968

MILITARY SPECIFICATION

PRIMER, NO. 1-1/2

1. SCOPE

1.1 This specification covers one type of fuze primer.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood-Preservative; Water-Repellant.

MILITARY

MIL-B-2427 - Boxes, Ammunition Packing; Wood, Nailed.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering for Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL22-35-27 - Primer, No. 1-1/2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1375

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

AGENT T. C. GEORGE'S TARIFF

(Application for copies should be addressed to T. C. George, Agent, 2 Pennsylvania Plaza, New York, New York 10001).

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials used in the manufacture of the primer shall be as specified in Drawing DL22-35-27 and subsidiary drawings.

3.1.1.1 Primer mixture. The primer mixture shall be uniform, non-mercuric, and noncorrosive. Diazodinitrophenol (dinol) shall be the primary explosive.

3.1.2 Components. All components of the primer shall conform to the specifications and drawings cited on Drawing DL22-35-27 and subsidiary drawings.

3.2 Assembly. The primer shall be assembled as shown on Drawing C22-35-27.

3.3 Stability. The primer shall remain stable for no less than one year at 85 percent relative humidity and temperatures of 115° to 160° F.

3.4 Sensitivity. All primers in a group shall fire at 19 inches or less, and all primers in a group shall misfire at 2 inches or more

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when drop tested as specified in 4.2.4. A "fire" is a primer ignition, regardless of whether or not there is an audible report on impact of the firing pin. H average plus five times the standard deviation ($\bar{H}$ + 5s) shall be no greater than 19 inches, when calculated as specified in 4.2.4.2.

3.5 Workmanship. The primer shall be free of oil, and grease and damage, such as dents, cracks, or deformation, which could affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of the primers of one interfix number produced by one manufacturer, from one manufacturing unit composed of charger and press operator, from the same materials, during the same daily work shift, for no more than five consecutive days.

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. A sample of 475 primers shall be randomly taken from each lot. If the lot represented is accepted, unused primers in the sample may be returned to the lot.

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4.2.3 Inspection procedure.

4.2.3.1 For examination. Sample primers shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For test. Sample primers shall be tested in accordance with 4.2.4. If misfires occur at 19 inches, the lot shall be rejected. If fires occur at 2 inches, the lot shall be rejected. If $\bar{H} + 5s$ is greater than 19 inches, the lot shall be rejected.

4.2.3.3 Classification of defects.

(a) Primer, No. 1-1/2, assembly (Dwg C22-35-27).

Categories	Defects
Critical:	None defined
Major:	AQL 0.40 percent defective
101	Diameter incorrect
102	Depth incorrect
103	Workmanship (3.5)

4.2.4 Sensitivity test. The primer sensitivity shall be determined by a drop test, using a test fixture consisting of a steel die to hold the primer, a firing pin having a spherical radius of 0.20 inch, a steel ball drop weight weighing 1.94 ± 0.02 ounce, and supports as necessary to control the fixture (see figure 1). The test shall be conducted as follows:

4.2.4.1 Procedure. Insert the primer into the die by hand, adjusting it so that all blows lie within 0.02 inch of the center of the primer being tested. Divide the sample of 475 primers into 17 groups of 25 each and one group of 50. Test the first group of 25 primers by dropping the steel ball from a height of 10 inches, measured as the distance between the top of the firing pin and the bottom of the ball. Record the number of misfires which occur [see 4.2.4.2(a)]. Continue testing groups of 25 primers at one-inch increments until a height is reached at which all primers in a group fire. If the height at which all primers fire is 19 inches or less, continue testing groups of primers, dropping the steel ball from a height of 9 inches, and then at one-inch decrements, until a height is reached at which all primers in a group misfire. If a fire or fires occurs at 3 inches, test the group of 50 primers at 2 inches.

4.2.4.2 Calculations. Calculate H average ($\bar{H}$) and the standard deviation (s) as follows:

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(a) Tabulate the data from the test into columns as shown (for example) in table I.

Table I. Primer sensitivity test

Title Number	1 Height of drop (inches)	2 Number of misfires	3 Fraction misfired	4 Correction factor	5 Column (3)(4)
	10	24	0.96	1	0.96
	11	18	0.72	3	2.16
	12	13	0.52	5	2.60
	13	7	0.28	7	1.96
	14	1	0.04	9	.36

(1) In column 1, list all drop heights in order, starting at the lowest height. DO NOT list the heights at which all primers fired or misfired.

(2) In column 2, list the number of primers which misfired at each height listed in column 1.

(3) In column 3, list the figure derivied by dividing the number of misfires recorded in column 2 by the number of primers in the group tested at the height in column 1. Calculations shall be recorded to the nearest one-hundredth (.01).

(4) In column 4, list odd numerals in sequence, beginning with 1, for as many drop heights as are listed in cloumn 1.

(5) In column 5, list the results obtained by multiplying the entries in columns 3 and 4.

(b) Use the following formula to calculate H average:

$$H = A+B+C$$

where: A = the sum of column 3,
B = the first drop height at which all of a group of primers misfire, and
C = one-half the drop-height increment.

(c) Use the following formula to calculate the standard deviation:

$$s = D\left(\frac{E}{} - F\right)$$

where: D = the drop height increment,
E = the sum of column 5, and
F = the square of the sum of column 3.

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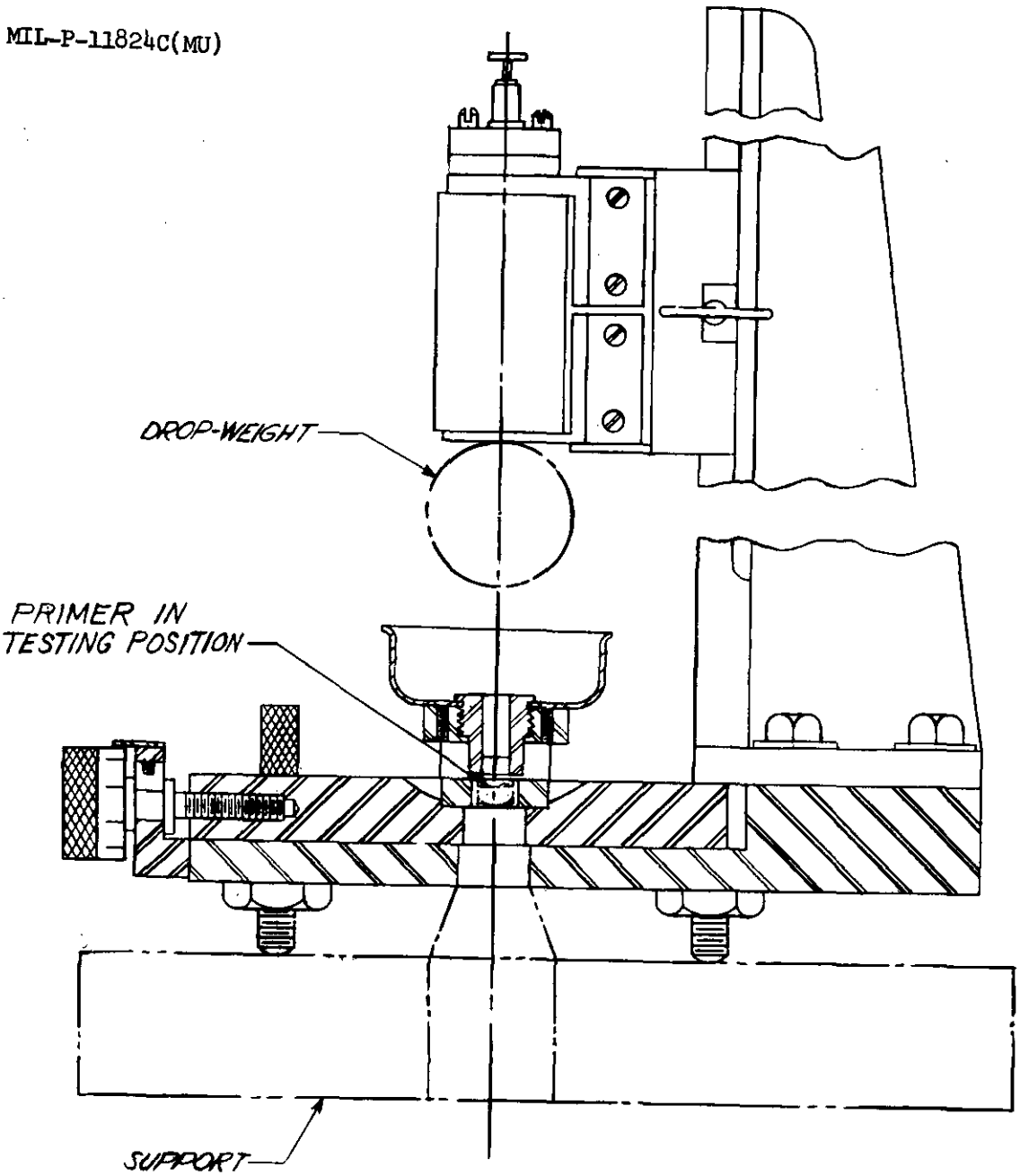


Figure 1.-Fixture, testing, primer sensitivity.

FOR ASSEMBLY SEE ORDNANCE
CORPS DRAWING NO. 81-3-44

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(d) Substituting the data used as examples in table I, the value of $\bar{H} + 5s$ would be calculated as follows:

$$\bar{H} = A+B+C$$
$$\bar{H} = 2.52 + 9.0 + 0.5$$
$$\bar{H} = 12.02$$

$$s = D\left(\frac{E-F}{}\right)$$
$$s = 1\left(\frac{8.04-6.35}{}\right)$$
$$s = 1.30$$

$$\bar{H} + 5s = 12.02 + 5(1.30)$$
$$\bar{H} + 5s = 18.52$$

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2).

5.1.1 Levels A and C. Primers shall be packaged in accordance with Drawing C22-35-37.

5.1.2 Interplant shipment (see 5.2.3).

5.2 Packing (see 6.2).

5.2.1 Level A. Primers shall be packed in accordance with Drawing D22-35-36, except the shipping container and all wood parts thereof shall be fully immersed for a minimum of three minutes in a water repellent, wood preservative conforming to composition A of TT-W-572. Prior to packing the primers in the treated shipping containers, the preservative shall be dry as evidenced by the dryness test specified in MIL-B-2427.

5.2.2 Level C. Primers shall be packed in accordance with Drawing D22-35-36.

5.2.3 Interplant shipment (see 6.3). "A designated quantity" of primers, not to exceed the weight limitations specified as follows, shall be packaged and packed in containers or special packs as applicable, together with interior packaging constructed as to give assurance that the primers being shipped will remain in prime condition and adequate to assure safe delivery to destination without damage to contents. The containers shall comply with Uniform Freight Classification, National Motor Freight Classification, or Agent T. C. George's Tariff, as applicable, and be acceptable for shipment at the most favorable rate. The containers shall be sufficiently rigid to withstand transportation and permit storing in tiers at least 10 feet high without damage to containers or contents. The gross weight and size limitations of the containers shall not exceed the tariff or classification limitations specified therein, and the containers shall be closed or strapped as specified therein.

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5.3 Marking.

5.3.1 Levels A and C. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with Drawing D22-35-39; shipping containers shall be marked in accordance with Drawing D22-35-36.

5.3.2 Interplant shipment (see 6.3). In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the domestic shipping requirements as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The primer is a component of one type of fuse used in floating smoke pots and some grenades.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Special marking, if required.
(c) Level of packaging and packing.

6.3 Interplant shipment. Packaging that does not enter the military supply system. Typical are shipments from a vendor to a contractor, a subcontractor or a supplier to a military arsenal or plant.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1375-A078

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